

B-Tree Insertions

B-Trees, Multiway Search Trees, Node Splitting · Mentor-Led Lab (Student Edition)

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Difficulty ★★★★★ Advanced

Problem

Pattern: multiway balanced tree with proactive splitting. Insert a sequence of keys into a B-tree of minimum degree t (each node holds up to $2t-1$ keys), splitting full nodes on the way down.

Return the tree printed level by level and its height in edges.

What to Compute

- Level i : the nodes on level i , each node's keys comma-separated and nodes space-separated.
- Height: the height in edges (a single-node tree is 0).

Input / Output Format

Input: Line 1: t (minimum degree). Line 2: N . Line 3: N integer keys.

Output: One Level i line per level, then Height.

Examples

Example 1

Input: 2 10 10 20 5 6 12 30 7 17 3 15

Output:

```
Level 0: 10
Level 1: 6 20
Level 2: 3,5 7 12,15,17 30
Height: 2
```

Explanation: With $t=2$ (max 3 keys per node), the inserts split nodes until the root holds 10 over two internal nodes and four leaves, height 2.

Example 2

Input: 2 7 1 2 3 4 5 6 7

Output:

```
Level 0: 2,4
Level 1: 1 3 5,6,7
```

Height: 1

Explanation: Sequential inserts split repeatedly so the root holds the separators 2 and 4 above three leaves, height 1.

Constraints

- Each node holds at most $2t-1$ keys and splits when full.
- Split a full child before descending into it (proactive splitting).
- Keys within a node stay sorted.

Sample Run

Sample Input

```
2
10
10 20 5 6 12 30 7 17 3 15
```

Sample Output

```
Level 0: 10
Level 1: 6 20
Level 2: 3,5 7 12,15,17 30
Height: 2
```